Toolkit: Chemistry of the Earth

Practical Eight: Stable isotopes

Objectives:

- Understand the terminology of stable isotope studies
- Investigate how stable isotope geothermometers are used to interpret formation of rocks and mineral deposits

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FEEDBACK

Stable isotopes for H, C, N, O, S:

Element	Isotope	Relative abundance (%)
Hydrogen	^{1}H (protium)	99.9844
	^{2}H (D)	0.0156
Carbon	^{12}C	98.89
	^{13}C	1.11
Nitrogen	^{14}N	99.63
	^{15}N	0.37
Oxygen	^{16}O	99.763
	^{17}O	0.0375
	^{18}O	0.1995
Sulfur	^{32}S	95.02
	^{33}S	0.75
	^{34}S	4.21
	^{36}S	0.02

A. Stable isotope terminology

1. **Isotopologues** are molecules that differ from each other only in isotopic composition.

a) List the isotopologues of oxygen (O_2), along with their masses:

	$^{16}O^{16}O$	$^{16}O^{17}O$	$^{16}O^{18}O$	$^{17}O^{18}O$	$^{17}O^{17}O$	$^{18}O^{18}O$
mass	32	33	34	35	34	36

b) List the isotopologues of water (H_2O), along with their masses: (use the symbols H for ^{1}H and D for ^{2}H)

	$H_2{}^{16}O$	$H_2{}^{17}O$	$H_2{}^{18}O$	$HD^{16}O$	$HD^{17}O$	$HD^{18}O$	$D_2{}^{16}O$	$D_2{}^{17}O$	$D_2{}^{18}O$
mass	18	19	20	19	20	21	20	21	22

2. The **Fractionation Factor (α)** describes the distribution of stable isotopes between two phases:

$\alpha_{A\text{-}B} = R_A / R_B$ where R is the ratio of the heavy to light isotope

Example: For oxygen isotope fractionation between quartz and magnetite,

$$\alpha_{qz\text{-}mt} = (^{18}O/^{16}O)_{quartz} / (^{18}O/^{16}O)_{magnetite}$$

a) Write the α equation for fractionation of the sulphur isotopes ^{32}S and ^{34}S, between sphalerite (sph) and galena (gal):

$$\alpha_{sph\text{-}gal} = (^{34}S/^{32}S)_{sphalerite} / (^{34}S/^{32}S)_{galena}$$

b) Instead of α, we commonly use the term **1000lnα**. This is because $1000\ln(1.00X) \approx X$.

(Another way to think of X is that X = (α-1)*1000.)

Show that the above relationship is true:

i) For sulphur isotope exchange between sphalerite and galena, the fractionation factor $\alpha_{(sph-gal)}$ is 1.00360 at 200 °C.

What is the value of X we are considering? X is 3.60, i.e. the 3rd and successive decimal places of α.

Another way to think of X is that it is (α-1)*1000.

Calculate the value of 1000lnα: 1000lnα = 1000ln(1.00360) = 3.594

ii) For oxygen isotopic exchange between quartz and calcite, the fractionation factor $\alpha_{(qz-cc)}$ is 1.01017 at 500 °C.

What is the value of X we are considering? In this case, X is 10.17.

Calculate the value of 1000lnα: 1000lnα = 1000ln(1.01017) = 10.12

3. **Delta notation, δ** is used for comparing an isotope ratio to the ratio measured in a standard.

$\delta = (R_{sample} - R_{standard}) / R_{standard} * 1000$ ‰

For oxygen isotopes (^{18}O and ^{16}O), the expression is:

$\delta^{18}O = (({}^{18}O/{}^{16}O)_{sample} - ({}^{18}O/{}^{16}O)_{SMOW}) / ({}^{18}O/{}^{16}O)_{SMOW}) * 1000$ ‰

(SMOW = Standard Mean Ocean Water, a universal standard for O and H)

a) Write the delta expression for sulphur isotopes ^{32}S and ^{34}S. The standard for sulphur isotopes is the mineral troilite (FeS) in the iron meteorite, Canyon Diablo (the one that made Meteor Crater, Arizona). The abbreviation for this is CDT (Canyon Diablo troilite). Remember to include units.

$\delta^{34}S = (({}^{34}S/{}^{32}S)_{sample} - ({}^{34}S/{}^{32}S)_{CDT}) / ({}^{34}S/{}^{32}S)_{CDT}) * 1000$ ‰

b) Write the delta expression for hydrogen isotopes, H and D. The standard for H is SMOW (see above). Remember to include units.

$\delta D = ((D/H)_{sample} - (D/H)_{SMOW}) / (D/H)_{SMOW}) * 1000$ ‰

4. **Combining α and δ** gives the following equalities:

$$10^3 \ln\alpha \approx \delta_A - \delta_B = \Delta_{A\text{-}B}$$

a) Consider the equilibrium between calcite (cc) and carbon dioxide (CO_2) in the atmosphere, at 20 °C. If we measure $\delta^{13}C$ in both these phases, we find that $\delta^{13}C_{CC}$ = +3.12 ‰ and $\delta^{13}C_{CO2}$ = -7.0 ‰.

Calculate the value of $\Delta_{CC\text{-}CO2}$: $\Delta_{CC\text{-}CO2} = 3.12 - (-7.0) = 10.12$

Now calculate the value of the fractionation factor, α : (Remember that $e^{(\ln X)} = X$)

$1000 \ln \alpha = 10.12$

$\ln \alpha = 10.12/1000 = 0.01012$

$\alpha = e^{0.01012} = 1.01017$

b) Burning fossil fuels changes the value of $\delta^{13}C_{CO2}$ in the atmosphere. If the value of $\delta^{13}C_{CO2}$ decreases to -12 ‰ as a result of burning fossil fuels, what would be the value of $\delta^{13}C$ for calcite that precipitates in equilibrium with atmospheric CO_2? (At the same temperature, 20 °C)

$1000 \ln \alpha = \Delta_{CC\text{-}CO2} = 10.12$ Remember to include units.

$10.12 = \delta^{13}C_{CC} - (-12)$ so $\delta^{13}C_{CC} = 10.12 - 12 = -1.88$ ‰

5. This example shows how we can **use the relationship between α and δ to determine the $\delta^{18}O$ value** for a given sample of water.

In the laboratory, we equilibrate the water with CO_2 gas at 25 °C. Then we extract the CO_2 gas into a mass spectrometer and measure its $\delta^{18}O$ value. It is easier to measure the isotopic composition of a gas than the isotopic composition of a liquid.

We measure $\delta^{18}O$ for CO_2 = +10 ‰.

The fractionation factor $\alpha_{(CO2)-(H2O)}$ is known: $\alpha_{(CO2)-(H2O)}$ = 1.04120 at 25 °C.

Calculate: 1000 ln $\alpha_{(CO2)-(H2O)}$ = 40.37

Now use the relationship:

1000 ln $\alpha_{(CO2)-(H2O)}$ = $\Delta_{(CO2)-(H2O)}$ = $\delta_{(CO2)} - \delta_{(H2O)}$ to determine $\delta_{(H2O)}$.

$40.37 = 10 - \delta_{(H2O)}$

$\delta_{(H2O)} = 10 - 40.37 = -30.37$ ‰

The $\delta^{18}O$ value of the water is: -30.37 ‰ (include units)

B. Stable isotope geothermometry: Determining the temperature of equilibration of two phases using their stable isotope compositions.

1. Oxygen isotope equilibria: We will determine the oxygen isotope composition of muscovite in a rock that equilibrated with water. The water has a known $\delta^{18}O$ value and the temperature of the reaction is known. (This is taken from Question 6 of Misra, p. 278.)

Fractionation of oxygen isotopes between muscovite and water in the temperature range 500 – 800 °C is described by the equation:

$1000\ln \alpha_{musc\text{-}H2O} = ((1.90 * 10^6)/T^2) - 3.10$

If the water has $\delta^{18}O$ = -9 ‰, and the temperature is 500 °C, what is the value of $\delta^{18}O$ for the muscovite? (Remember that T must be in Kelvin)

T = 500 + 273 = 773 K

$1000\ln \alpha_{musc\text{-}H2O} = ((1.90 * 10^6) / 773^2) - 3.10 = 0.0798$

$1000\ln \alpha_{musc\text{-}H2O} = \Delta_{musc-H2O} = \delta_{musc} - \delta_{H2O}$

$0.0798 = \delta_{musc} - (-9)$

$\delta_{musc} = 0.0798 - 9 = -8.9202$ ‰

Remember to include units.

2. Sulphur isotope geothermometry: Sulphur isotopes are used to determine the temperature of formation of sulphide ore deposits. We will use the sphalerite (ZnS) – galena (PbS) sulphur isotope system to determine the temperature at which the Providencia, Mexico sulphide deposit formed. The data are from a paper by Rye (1974). (This is taken from Question 13 of Misra, p. 279, and Question 13.4 in McSween, p. 284.)

The sphalerite – galena geothermometer is defined by the equation:

$$T = 850 / \sqrt{(\Delta^{34}S_{sphal\text{-}gal})} \qquad \text{(T in Kelvin)}$$

a) Write the equation that relates $\Delta^{34}S_{sphal\text{-}gal}$ to $\delta^{34}S_{sphalerite}$ and $\delta^{34}S_{galena}$:

$$\Delta^{34}S_{sphal\text{-}gal} = \delta^{34}S_{sphalerite} - \delta^{34}S_{galena}$$

b) Complete the Table below to determine the temperature recorded by each sample:

Table 1: Sphalerite - galena geothermometer

Sample	$\delta^{34}S$ (sphalerite)	$\delta^{34}S$ (galena)	$\Delta_{sph\text{-}gal}$	T (K)	T (°C)
1	0.95	-1.15	2.10	587	314
2	0.87	-1.11	1.98	604	331
3	1.03	-1.42	2.45	543	270
4	0.01	-1.71	1.72	648	375

As an example, for sample 2:

$$\Delta^{34}S_{sphal\text{-}gal} = \delta^{34}S_{sphalerite} - \delta^{34}S_{galena}$$

$$\Delta^{34}S_{sphal\text{-}gal} = 0.87 - (-1.11) = 1.98$$

$$T = 850 / \sqrt{(\Delta^{34}S_{sphal\text{-}gal})}$$

$$T = 850 / \sqrt{1.98} = 604 \text{ K} = (604 - 273) \text{ °C} = 331 \text{ °C}$$

c) What is the temperature of formation of the lead-zinc sulphide deposit? How might we interpret the different temperatures given by different samples?

The temperatures recorded by different samples vary between 270 and 375 °C. Differences might arise because the temperature does truly vary across the deposit. Temperature might vary with space and / or time.

Also, there is a sampling problem. It is difficult to select samples that represent equilibrium between the two minerals. The sphalerite crystals analyzed were strongly zoned because they grew in successive generations of fluid. The paper by Rye (1974) discusses the problems selecting sphalerite and galena samples that represent equilibrium pairs. So there could be some uncertainty because of sampling.

There is an uncertainty associated with measuring the values of $\delta^{34}S$ in the laboratory. The Rye paper states that "the uncertainty of calculated temperatures due to uncertainty for $\Delta^{34}S_{sphal\text{-}gal}$ values (±0.1 ‰) is about ±15°C at 350°C." The measured temperature range is much greater than this error.

We could take the average of all four analyses and say that the formation temperature is 322 °C, with a standard deviation of 43. However, the number of analyses is small so taking an average is not very meaningful. Overall, it should be clear that just measuring one mineral pair is not enough.

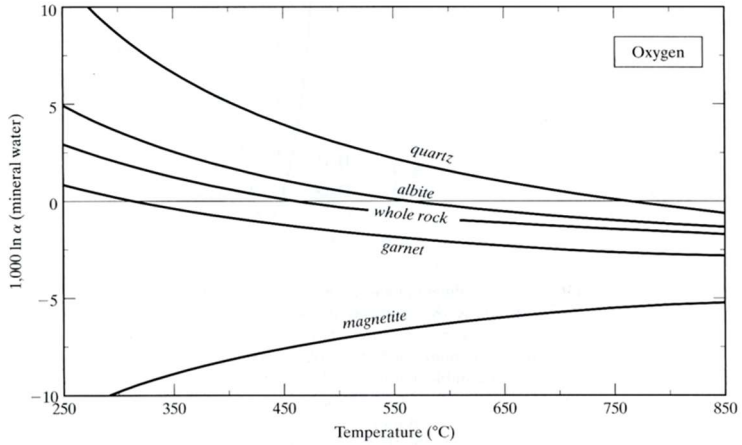
3. Oxygen isotope geothermometry: We will determine the temperature of equilibration of a gneiss (a metamorphic rock) that contains the minerals quartz and magnetite. (This is taken from Example 9.2 of White, p. 386.)

a) The figure below shows how oxygen isotopes fractionate between minerals and water as a function of temperature. Highlight the curves that show 1000ln $\alpha_{qz\text{-}H2O}$ and 1000ln $\alpha_{mt\text{-}H2O}$.

b) The fractionation factor 1000ln $\alpha_{qz\text{-}mt}$ is equivalent to 1000ln $\alpha_{qz\text{-}H2O}$ - 1000ln $\alpha_{mt\text{-}H2O}$.

What is the approximate value of 1000ln $\alpha_{qz\text{-}mt}$ at 300 °C? (just use integers) 9 – (-10) = 19

What is the approximate value of 1000ln $\alpha_{qz\text{-}mt}$ at 800 °C? (just use integers) 0 – (-5) = 5



c) The equation for the curve that gives the fractionation factor $\alpha_{qz\text{-}mt}$ is:

1000 ln $\alpha_{qz\text{-}mt}$ = 6.29 x $10^6/T^2$ (T is in Kelvin) (see Table 9.3 of White)

$\delta^{18}O$ values for quartz and magnetite were measured and found to be:

Quartz, $\delta^{18}O$ = 11.1 ‰ Magnetite $\delta^{18}O$ = 1.9 ‰

Find the temperature of equilibration of the gneiss: (Hint – first calculate $\Delta_{qz\text{-}mt}$)

Give your answer in Kelvin and °C.

$\Delta_{qz\text{-}mt} = \delta_{qz} - \delta_{mt} = 11.1 - 1.9 = 9.2$

$\Delta_{qz\text{-}mt} = 1000 \ln \alpha_{qz\text{-}mt} = 6.29 \times 10^6/T^2$

$9.2 = 6.29 \times 10^6/T^2$

$T^2 = (6.29 \times 10^6) / 9.2 = 683{,}695$

$T = 827$ K = 554 °C

Rye, R. O. (1974): A Comparison of Sphalerite-Galena Sulfur Isotope Temperatures with Filling Temperatures of Fluid Inclusions. Economic Geology, Vol.69(1), p.26-32.
http://economicgeology.org/content/69/1/26